

SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Cardiovascular Biomechanics | Medical Device R&D | Finite Element Modeling

University of Kentucky · Lexington, KY | Sara.b@uky.edu | [LinkedIn](#) | [GitHub](#) | [ResearchGate](#) | [Website](#)
Cardiac Mechanics · Soft-Tissue Biomechanics · Nonlinear FEA · Material Validation · PV Analysis · Fibrosis · Python

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky focused on cardiovascular biomechanics and computational modeling. Research spans passive myocardial constitutive behavior, ventricular mechanics, fiber architecture, mechanical efficiency, baroreflex compensation, growth/remodeling, and fibrosis. Experienced in finite-element modeling, model validation, pressure-volume analysis, soft-tissue mechanics, Python-based data analysis, and translating physiological questions into quantitative engineering simulations.

EDUCATION

University of Kentucky — Lexington, KY Spring 2024–Present
PhD in Mechanical Engineering, Direct PhD Program · PhD Candidate since April 2026 · GPA: 3.91/4.00 · Advisor: Jonathan F. Wenk, PhD

Iran University of Science and Technology (IUST) — Tehran, Iran 2016–2021
B.Sc. in Mechanical Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant — Computational Cardiac Mechanics Spring 2024–Present
University of Kentucky · Advisor: Jonathan F. Wenk, PhD

- Develop and extend a coupled left-ventricular simulation framework integrating passive myocardial mechanics, active contraction, half-sarcomere kinetics, and cardiovascular circulation.
- Compared nonlinear anisotropic myocardial constitutive formulations to determine how material-law choice changes ventricular deformation, stress, pressure-volume behavior, and remodeling predictions.
- Developed material-validation workflows using biaxial, shear, and passive-inflation simulations to assess soft-tissue constitutive behavior before organ-level studies.
- Investigated myocardial fiber disarray and quantified resulting changes in pump function, mechanical efficiency, ATP-related energetics, and baroreflex compensation.
- Currently developing an agent-based fibrosis model coupled with finite-element cardiac mechanics to study how spatial fibrosis alters myocardial mechanics and ventricular function.
- Analyze pressure-volume loops, stress/strain distributions, fiber architecture, and spatial simulation outputs using Python, ParaView, and scientific post-processing workflows.

Engineering Data & Machine Learning

Additional research project

- Applied vision-language models and regression methods to electrospinning video data for prediction of fiber morphology.
- Built repeatability and cross-validation workflows to evaluate robustness of model-derived features and predictions.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. “Advancements in finite element modeling of myocardial infarction: a review.” *Biomechanics and Modeling in Mechanobiology*, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Manuscripts Under Review

- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions.” Manuscript under review.
- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation.” Manuscript under review.

Selected Conference Presentation

Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions.” Poster, Biomedical Engineering Society (BMES) Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant — Statics, University of Kentucky *Fall 2026*
Supported undergraduate instruction and problem solving in equilibrium, force/moment analysis, trusses, and frames.

Teaching Assistant — Statics, Iran University of Science and Technology *2021*

TECHNICAL SKILLS

- Biomechanics & R&D:** Cardiovascular mechanics, soft-tissue mechanics, anisotropic constitutive modeling, myocardial fiber architecture, fibrosis, growth/remodeling
- Simulation & Validation:** Nonlinear FEA, FEniCS/DOLFIN, material validation, biaxial/shear testing, passive inflation, pressure-volume analysis
- Programming & Analysis:** Python, MATLAB, NumPy, SciPy, Pandas, UFL
- Visualization:** ParaView, Matplotlib, HDF5, XDMF
- Engineering Software:** COMSOL Multiphysics, SolidWorks; HPC workflows with MPI/PETSc/SLURM